Object Detection for Aerial Images With Feature Enhancement and Soft Label Assignment

Ying Yu, Xi Yang, *Member, IEEE*, Jie Li, and Xinbo Gao, *Senior Member, IEEE*

***Abstract*— Object detection in aerial images, different from general object detection, faces several challenges, such as arbitrary-oriented objects and extremely imbalanced foreground–background distribution. Although some recent proposed aerial object detection methods achieve promising results, they are mainly anchor-based detectors that rely heavily on predefined anchor boxes, and the final detection performance is sensitive to anchor-related hyperparameters. In contrast, in this article, we present an anchor-free detector with feature enhancement and soft label assignment (FSDet), which adopts a simpler design and achieves competitive performance. Specifically, to address the feature misalignment for detecting oriented objects, we propose an oriented feature refinement (OFR) module to align the features with oriented objects. To alleviate the background issue, we design a class-aware context aggregation module to integrate the intraclass context information and suppress the background context. Moreover, we propose a soft label assignment mechanism to measure the weight of training samples within the arbitrary-oriented objects, which can concentrate more on representative items with regard to their potential to detect oriented objects, achieving a more stable optimization during training. Extensive experiments on several datasets suggest that the proposed method is superior to the state-of-the-art methods and achieves a better tradeoff between speed and accuracy.**

***Index Terms*— Aerial images, anchor-free detector, class-aware context aggregation (CCA), oriented feature refinement (OFR), soft label assignment (SLA).**

I. Introduction

OBJECT detection in aerial images has drawn great attention recently, which aims to localize the objects in images and identify their categories simultaneously [1]–[5]. It can also bring benefits to numerous applications in both civilian and military domains, such as urban management, environmental monitoring, and military investigation.

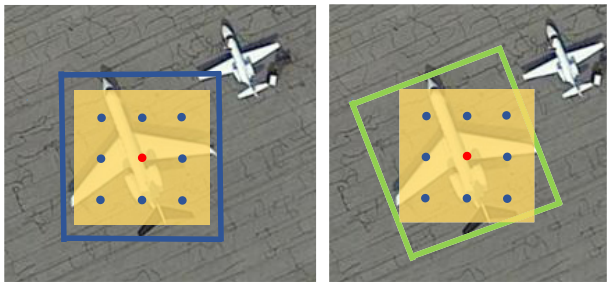


Fig. 1. Illustration of misalignment between the receptive field of convolutional feature (yellow regions) and oriented box. (Left) Horizontal box regression. (Right) Oriented box regression. The blue rectangle is the horizontal box, and the green rectangle is the oriented box.

Currently, a large number of general object detection methods [6]–[10] have been proposed and achieved promising performance in natural image datasets, which have greatly promoted the development of object detection in aerial images. In general, these methods can be divided into two classes, namely, anchor-based and anchor-free. Anchor-based methods [2], [3], [5], [11] preset a lot of anchor boxes and then predict the offsets to decode the final detections. In contrast, anchor-free methods [12]–[14] remove the setting of anchors and regress the distance to bounding boxes directly. Although the performance of anchor-free detectors is not as satisfactory as state-of-the-art anchor-based methods in aerial images, they are typically more flexible and can be easily generalized to various scenarios with a much simpler network structure that eliminates the predefined anchor boxes and avoids anchor-related hyperparameters. In this work, we focus on anchor-free detectors for object detection in aerial images, aiming to remove the obstacle impeding them from achieving promising results and maintain their strengths of simple structure and high efficiency at the same time.

In practice, object detection in aerial images is more challenging than in natural scenes in two aspects. On the one hand, the objects are usually arranged in arbitrary orientations due to the bird's-eye view perspective. Thus, the oriented bounding box (OBB) regression is widely used to detect arbitrary-oriented objects, which also increases the difficulty of accurate localization. As illustrated in Fig. 1, unlike horizontal box regression, the axis-aligned receptive field of standard 2-D convolution is not well aligned with the oriented box

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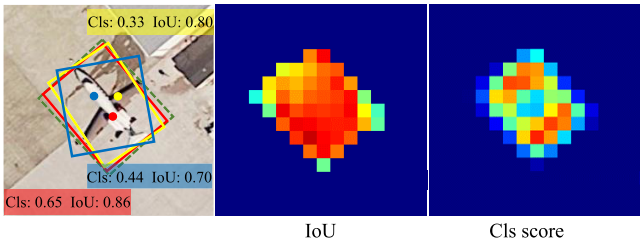


Fig. 2. Illustration of inconsistent detection at different locations. The green dotted box denotes the ground truth, and the other three boxes are the detection results of FCOS [15] with a rotational head at corresponding points. The "IoU" and "Cls score" columns denote the predicted IoU and classification score of different locations, respectively.

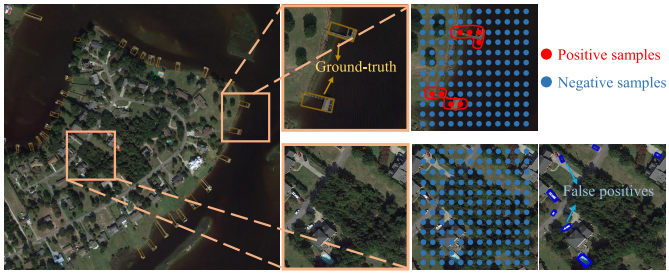


Fig. 3. Illustration of imbalanced foreground–background distribution in aerial images. This leads to an imbalance between the number of positive and negative samples, making it hard to learn a strong and unbiased classifier. On the other hand, some background area in aerial images is complex and hard to distinguish, which may cause some false positives, and we refer to this as background interference.

exactly considering the object orientation and shape; thus, the convolutional features cannot accurately and clearly describe the information of the oriented object. Moreover, we observe that positive samples at different spatial locations exhibit inconsistent detection ability. For example, as depicted in Fig. 2, the prediction accuracy of these three samples at different locations is distinct. On the other hand, the ratio of foreground area is much less than in natural images, which exacerbates the imbalance between the number of positive and negative samples, as shown in Fig. 3, making it hard to learn a strong classifier.

In this article, to tackle the aforementioned problems, we design a new anchor-free detector with feature enhancement and soft label assignment (FSDet). The proposed FSDet contains two feature enhancement modules that extract more effective features for oriented objects in aerial images and a soft label assignment (SLA) strategy for network training. Specifically, to address the feature misalignment for detecting oriented objects, we propose an oriented feature refinement (OFR) module to align features with the objects by utilizing the geometric information of oriented boxes. Second, to alleviate the background interference caused by the foreground–background imbalance, we design a class-aware context aggregation (CCA) module to enhance the discrimination of foreground intraclass features by explicitly foreground modeling. The aggregation module fully exploits the context information of the same class and suppresses a large number of background pixels and the information of other irrelevant classes. Third, to deal with the inconsistency of samples at different locations during training, we propose an SLA strategy to assign soft weights to the samples within the arbitrary-oriented objects with different detection abilities, thus considering the samples distinctly and concentrating more on representative items.

We conduct experiments on several datasets for object detection in aerial images, such as DOTA1.0, DOTA1.5, HRSC2016, and UCAS-AOD. The results show that the proposed method is superior to previous anchor-free methods and attains competitive detection precision with state-of-the-art anchor-based detectors.

Our contributions can be summarized as follows.

1) To address the feature misalignment for detecting oriented objects, an OFR module is proposed to extract more effective features by utilizing the geometric information of oriented boxes.
2) To alleviate the overintroduced background and learn more discriminative foreground class features, a lightweight CCA module is designed to fully exploit the context information of a specific class and suppress the effect of background and other classes.
3) To resolve the inaccurate label assignment in arbitrary-oriented object detection, an SLA strategy is proposed to treat the samples with different detection abilities distinctively, leading to more stable network training and accurate prediction.

II. Related Work

A. Object Detection in Aerial Images

Object detection in aerial images has received increasing attention in recent years. The great process of generic object detection methods adopting a convolutional neural network (CNN) has strongly promoted research in this field. A big difference between aerial images and natural images is that the objects in aerial images are often distributed in arbitrary directions. Therefore, a lot of methods have been proposed for this question and achieved outstanding performance. These methods can be categorized into two classes, i.e., anchor-based detectors and anchor-free detectors. Anchor-based methods preset a large number of anchors at each location and then predict the offsets between ground-truth boxes and anchors to get the final results. For example, RRPN [11] is built on the classical two-stage detection framework, i.e., Faster R-CNN [8], and adopts rotated anchors in the rotation region proposal network to generate inclined proposals that are more fit with the oriented boxes; then, it proposes a rotation region-of-interest (RoI) pooling layer to extract the rotation-invariant features from the rotational regions. RRD [16] treats classification and regression tasks separately considering their different characteristics, that is, employ rotation-invariant features for classification and rotation-sensitive features for regression to achieve more precise localization. After that, the RoI transformer [2] proposes a more efficient rotated RoI learner to transform the horizontal RoI into rotational RoI and then predicts the oriented boxes based on the inclined region proposals. However, these methods are computationally expensive and,

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therefore, take a long time during inference. Later, to attain high detection accuracy with a fast inference speed, R^3Det [4] designs a refined one-stage detector that adopts a feature refinement module to address the feature misalignment problem during the refinement stage. Compared with these anchor-based methods, anchor-free detectors have a more simplified architecture that predicts the bounding boxes directly without any anchors. IENet [12] is a one-stage anchor-free detector built on FCOS [15] network and uses an independent branch to regress the orientation parameter. Axis learning [13] detects the objects by predicting their axis and width at each pixel of feature maps directly. Later, BBAVectors [14] is extended from the keypoint-based object detector CenterNet [17] that designs a group of box boundary-aware vectors to locate the OBBs instead of predicting the angle directly.

B. Feature Alignment in Object Detection

Previous studies [18]–[20] have shown that the misalignment between the convolutional features and anchors or RoIs affects the detection performance. Two-stage detectors employ the RoI pooling or RoI align operation to extract features from the RoI regions. To detect oriented objects, Ding *et al.* [2] propose a rotated position-sensitive RoI align layer to obtain features aligned with the oriented proposals. In one-stage detectors, Cascade RetinaNet [18] adopts deformable convolution [21] to address the feature misalignment, and the offsets of deformable convolution are learned during training. AlignDet [19] also uses deformable convolution, but its offsets are computed by the refined anchors. For oriented object detection, Han *et al.* [20] propose an alignment convolution to alleviate the misalignment between the features and arbitrary-oriented objects. In this work, we employ an OFR module to align features with the objects by utilizing the geometric information of oriented boxes.

C. Feature Representation Learning

Object detection in aerial images suffers from severe background interference caused by the imbalanced foreground–background distribution. Some attempts try to alleviate this problem by employing the attention mechanism [3], [22] to suppress the noisy features and highlight the effective features. For example, SCRDet [22] presents a supervised pixel attention and channel attention network to concentrate more on the object regions. CAD-Net [3] designs a spatial-and-scale-aware attention module to guide the network to focus on more appropriate spatial locations and feature scales. Some other methods make use of context information for capturing more robust features [23], [24]. However, these methods fail to incorporate the classes into the context. Besides, there are a few works that employ contrastive learning aiming to pull together the feature embedding of samples belonging to the same class and push apart feature embedding of samples from different classes [25], [26]. Our CCA module can fully exploit the useful intraclass context information and suppress the large amounts of noise simultaneously; thus, the features of the same class are more compact, and features of different classes are more separated.

D. Label Assignment in Object Detection

The selection of positive and negative samples has a great influence on network training in object detection. Traditional anchor-based methods [2], [3], [5], [11] assign the labels to anchors according to the intersection-over-union (IoU) between anchors and ground-truth boxes. Specifically, the anchors with IoU greater than a threshold with any ground-truth box are selected as positive samples, while the others are regarded as negative samples or ignored. As for anchor-free detectors [12]–[14], a common approach is to treat the locations inside the original or shrunk ground-truth boxes as positive samples. However, this kind of label assignment relying on heuristic predefined rules may not be accurate enough, thus introducing label noise during the training process. Some methods have been proposed to solve this problem. FreeAnchor [27] optimizes the object-anchor matching through a maximum likelihood estimation procedure and learns to select the most representative anchors from a bag of anchors for each object dynamically. After that, ATSS [28] selects the training samples adaptively by adjusting the IoU threshold according to the statistical characteristics of objects. On the other hand, PAA [29] defines the positive and negative samples from the view of probability distribution by maximizing the likelihood of anchor scores that are related to the model's learning status. Later, DAL [30] introduces a superior matching degree than IoU to measure the localization potential of anchors and then assigns the labels to anchors based on this newly defined metric. AutoAssign [31] proposes a fully differentiable label assignment mechanism, which considers the selection of training samples in spatial and scale dimensions simultaneously. GGHL [32] proposes a label assignment strategy for arbitrary-oriented objects that defines the positive samples based on a 2-D Gaussian heat map. FCOSR [33] also focuses on the label assignment of oriented objects, which employs ellipse center sampling, fuzzy sample label assignment, and multilevel sampling to assign labels to training samples more appropriately. Inspired by this excellent work, we extend the idea of dynamic label assignment from general object detection to aerial object detection tasks. Different from the above methods, the proposed SLA strategy assigns soft weights to the samples adaptively, so as to treat them distinctively according to their quality. Moreover, our method takes into account the instances with arbitrary directions to better adapt to the aerial scenarios.

III. METHODOLOGY

In this section, we introduce the proposed FSDet in detail. As illustrated in Fig. 4, our FSDet is built on the structure of classical one-stage anchor-free detection network, FCOS [15]. We design an OFR module and a CCA module to obtain more effective features. Moreover, the SLA strategy is utilized to address the inaccurate label assignment for positive and negative samples during network training.

A. Baseline Network

We construct a one-stage anchor-free detector as the baseline network extended from FCOS. It consists of a backbone

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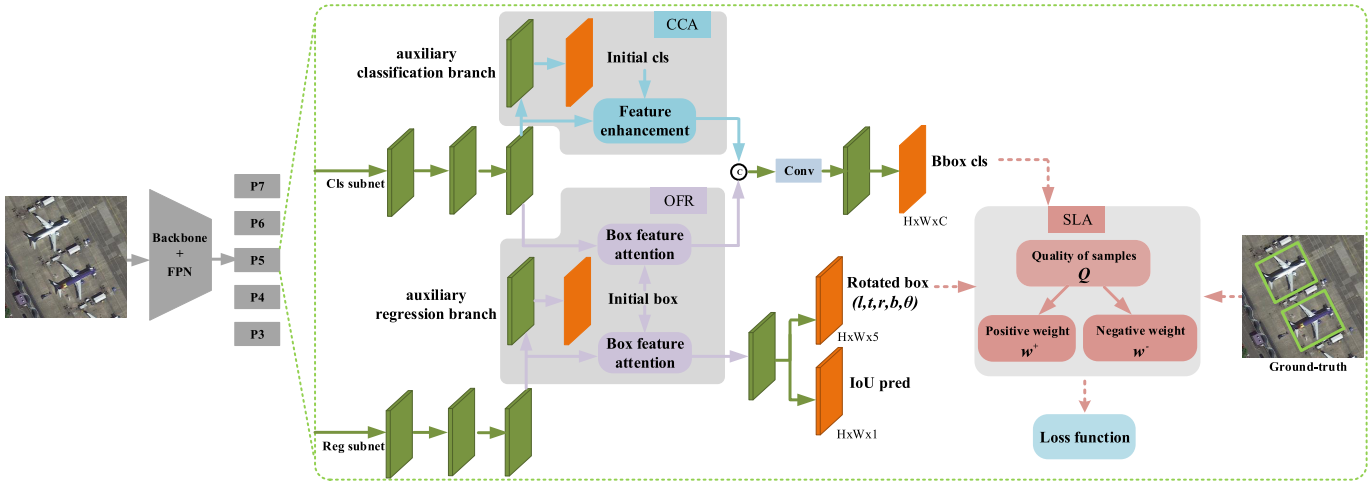


Fig. 4. Overview of the proposed FSDet. The FSDet is built on the FPN (P3-P7). The detection head in the original FCOS consists of two subnets for classification and regression, respectively. An IoU prediction branch is added to the regression subnet. The OFR module and the CCA module are designed for feature enhancement. For the classification subnet, the features from the OFR module and the CCA module are concatenated and then passed through a 1×1 convolutional layer for final prediction. SLA is employed during the training process to assign more appropriate attention to the samples with different localization abilities.

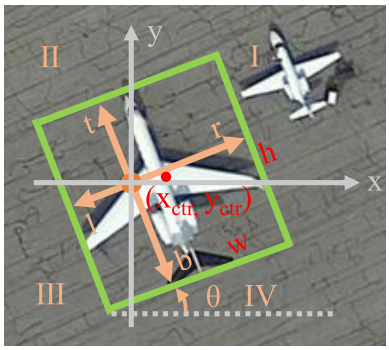


Fig. 5. Descriptions of OBB. $(x_{\text{ctr}}, y_{\text{ctr}}, w, h)$ are the center, width, and height of the OBB, respectively. The decoded OBB at each location is denoted by a 5-D vector (l, t, r, b, θ), where (l, t, r, b) represent the distance from the location to the left, top, right, and bottom sides of the OBB corresponding to the four quadrants of the Cartesian coordinate system, respectively. θ denotes the acute angle between the OBB and the x-axis, and the corresponding side is referred to as w.

network and two task-specific subnets for classification and regression. The feature pyramid network (FPN) [9] is adopted to extract multiscale features. The two subnets are fully convolutional networks with several stacked convolution layers. The classification subnet predicts the probability of objects at each location for each of the C object classes. The regression subnet predicts the distance from each location to the boundaries of a bounding box. For object detection in aerial images, the objects are usually arbitrary-oriented and annotated with OBBs, which can be represented as $B = (c, x_{\text{ctr}}, y_{\text{ctr}}, w, h, \theta)$, where $(x_{\text{ctr}}, y_{\text{ctr}})$ is the center and c, w, and h are the class label, width, and height of the OBB, respectively. $\theta \in [-90°, 0°)$ denotes the acute angle between the OBB and the x-axis, corresponding to the side w of OBB. Therefore, we add an additional angular offset to the regression subnet so that it can predict OBBs. As described in Fig. 5, we adopt a 5-D class-agnostic vector $d = (l, t, r, b, \theta)$ to decode the OBB at each location, where l, t, r, and b mean the offset from the location to the left, top, right, and bottom sides of the OBB distributed in four quadrants of the Cartesian coordinate system, respectively. θ is the angle of the OBB as mentioned before. In addition, we replace the centerness branch in the original FCOS with IoU prediction to predict the IoU between the regressed box and ground-truth box, which works better in subsequent ablation experiments.

During the training stage, the selection of positive and negative samples plays a crucial role in optimizing the network. Instead of anchor boxes in anchor-based detectors, we directly treat the locations on each level of the feature pyramid $\{P3, P4, P5, P6, P7\}$ as training samples. Each positive sample has a classification target c^*, and a regression target $d^* = (l^*, t^*, r^*, b^*, \theta^*)$. c^* is the class label of the associated ground-truth object, (l^*, t^*, r^*, b^*) is the offset from the location to four sides of the oriented box, and θ^* represents the acute angle of the oriented box. Because negative samples only participate in classification, they only have a classification target $c^* = 0$ corresponding to the background class. The other settings keep unchanged from the original FCOS.

Formally, for a positive sample located at (xs, ys) on feature map P_l, which is associated with a ground-truth box B, the corresponding location (x, y) on input image is denoted as

$$\begin{aligned} x &= \left\lfloor \frac{s}{2} \right\rfloor + xs \cdot s \\ y &= \left\lfloor \frac{s}{2} \right\rfloor + ys \cdot s \end{aligned} \tag{1}$$

where s indicates the stride of feature level l. Here, we adopt ResNet50 as the backbone network; hence, the stride s for the multiscale feature maps $P3$, $P4$, $P5$, $P6$, and $P7$ that are computed from the residual blocks of ResNet is set to 8, 16, 32, 64, and 128, respectively. Then, the regression target can

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be formulated as

$$l^* = \left(x - \left(x_{\text{ctr}} - \frac{w}{2}\right)\right)\Big/ s, \quad t^* = \left(y - \left(y_{\text{ctr}} - \frac{h}{2}\right)\right)\Big/ s$$
$$r^* = \left(\left(x_{\text{ctr}} + \frac{w}{2}\right) - x\right)\Big/ s, \quad b^* = \left(\left(y_{\text{ctr}} + \frac{h}{2}\right) - y\right)\Big/ s$$
$$\theta^* = \theta. \tag{2}$$

For the baseline network, the selection of positive and negative samples follows FCOS that samples inside a ground-truth box are positives, and all others are negatives; then, we can obtain a positive sample set S_n^+ for each ground-truth object n and a negative sample set S^-. Then, the multitask loss function for training the whole network is defined as

$$L = \frac{1}{N_{\text{pos}}} \sum_i \left(\sum_n \mathbb{1}_{\{i \in S_n^+\}} L_{\text{cls}}^+ (\hat{c}_i, c_i^*) + \mathbb{1}_{\{i \in S^-\}} L_{\text{cls}}^- (\hat{c}_i, 0) \right) + \frac{\lambda_1}{N_{\text{pos}}} \sum_i \sum_n \mathbb{1}_{\{i \in S_n^+\}} \left(L_{\text{reg}} (\hat{d}_i, d_i^*) + L_{\text{IoU}} (\hat{q}_i, q_i^*) \right) \tag{3}$$

where L_{cls}^+ and L_{cls}^- denote the classification losses for positive and negative samples, respectively. We adopt the same focal loss function in this article. i is the index of locations across all level of feature maps. $\hat{c}_i$ and c_i^* are the C-D classification output and the classification target of location i, respectively. $\hat{d}_i$ and d_i^* are the 5-D localization output and regression target of location i, respectively. q_i^* is the IoU target of location i, which is calculated as the IoU between predicted OBB and assigned ground-truth OBB. $\hat{q}_i$ is the predicted IoU of the network. N_{pos} represents the total number of positive samples. $\mathbb{1}_{\{\text{condition}\}}$ is the indicating function that equals to 1 if the condition is satisfied and 0 otherwise. L_{reg} represents the regression loss of positive samples defined as

$$L_{\text{reg}}(\hat{d}, d^*) = L_{\text{HBB}}((\hat{l}, \hat{t}, \hat{r}, \hat{b}), (l^*, t^*, r^*, b^*)) + \lambda_2 L_{\text{angle}}(\hat{\theta}, \theta^*). \tag{4}$$

L_{HBB} is the loss calculated using the horizontal bounding box (HBB) of the predicted OBB and ground-truth box, which adopts GIoU loss in [34], since it is scale-invariant to deal with objects of different scales. L_{angle} is the loss of rotation angle defined as

$$L_{\text{angle}}(\hat{\theta}, \theta^*) = 1 - \cos(\hat{\theta} - \theta^*). \tag{5}$$

L_{IoU} compares the values of predicted IoU from IoU prediction branch and the computed IoU between predicted OBB and ground-truth OBB. We adopt the binary cross-entropy loss for this function. Besides, λ_1 in (3) and λ_2 in (4) are hyperparameters that balance the weights of different loss terms and are set to 1 in our experiments.

B. Oriented Feature Refinement

The convolutional features with axis-aligned receptive fields are hard to represent the oriented objects accurately in accordance with their orientations and shapes; thus, to alleviate this issue, we propose an OFR module that utilizes the spatial geometric information of oriented boxes, aiming to learn more effective feature representation for oriented objects. This module consists of an auxiliary regression branch and a box feature attention mechanism, as illustrated in Fig. 4. The auxiliary regression branch outputs a 5-D vector (l, t, r, b, θ) for each location of the feature map that represents the initial oriented box. Based on this initial oriented box, the box feature attention mechanism is adopted to produce the refined feature maps that encode more accurate information of oriented objects for final predictions. We employ the box feature attention mechanism for the classification and regression subnets, respectively.

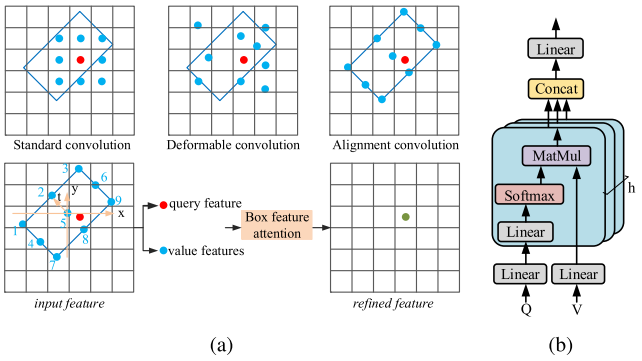


Fig. 6. (a) Illustration of different methods. (Top row) Sampling points in different methods with 3 × 3 kernels at a given location of the feature map. (Bottom row) OFR module, which takes the input feature as query feature and the features at nine fixed sampling points of the initial oriented box as value features and then produces the refined features. (b) Structure of the box feature attention mechanism.

Fig. 6(a) depicts the sampling points in different methods with 3 × 3 kernel at a given location of the feature map. The top row of Fig. 6(a) shows the sampling points of standard convolution with regular sampling locations, deformable convolution with deformable sampling locations, and alignment convolution with computed sampling locations, respectively. The bottom row depicts the proposed OFR module. At each location of the feature map, we predict an initial oriented box, and then, we map this box back to the feature map. The nine sampling points in alignment convolution and the proposed OFR module are the corresponding locations of the initial oriented box (including four vertices, the midpoints of the four edges, and the center point). As pointed out in previous studies [18]–[20], feature alignment is important for obtaining accurate detections. We can see that the sampling locations in standard convolution deviate from the oriented box; therefore, the extracted features cannot represent information about the objects accurately. Deformable convolution learns offsets to the sampling locations in standard convolution without explicit supervision; thus, there is no guarantee that the sampling points all fall on the objects. Alignment convolution in [20] extracts features from the fixed nine sampling locations of corresponding oriented boxes by utilizing deformable convolution. It alleviates the misalignment between axis-aligned convolutional features and arbitrary oriented objects, which adjusts the feature sampling locations with the guide of oriented boxes. However, the convolution operation in alignment convolution at different locations of the feature map shares the same 3 × 3 kernel.

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In this work, we design the box feature attention mechanism inspired by multihead attention in transformer [35]. As illustrated in Fig. 6, given a sampling location and an initial oriented box, the feature $\mathbf{f}_i^q \in \mathbb{R}^{d_f}$ at this location is treated as the query input, and the features at nine fixed sampling points of the oriented box are regarded as the value features $\mathbf{f}_i^v \in \mathbb{R}^{9\times d_f}$. The OBB is distributed in four quadrants of the Cartesian system, as described in Fig. 5; we assume an edge to be the top edge if the line segment from center point to this edge falls in the second quadrant, and then, we arrange the value features of these nine points clockwise from the top edge (see Fig. 6) to generate a sequence in order. Then, the box feature attention mechanism with h attention heads generates output feature $\mathbf{f}_i^o \in \mathbb{R}^{d_f}$ of the query $\mathbf{f}_i^q$ by calculating weighted average vectors of the value features $\mathbf{f}_i^v$

$$\begin{aligned}\mathbf{f}_i^o &= \text{Concat}\left(\text{head}_1^i, \ldots, \text{head}_h^i\right) W^O \\ \text{head}_k^i &= \sum_{j=[1,\ldots,9]} \text{softmax}\left(T_k\left(\mathbf{f}_i^q\right)\right)^* \left(\mathbf{f}_i^v {W_k}^V\right)\end{aligned} \tag{6}$$

where the projections matrices $W^O \in \mathbb{R}^{d_f\times d_f}$ and $W_k^V \in \mathbb{R}^{d_f\times d_v}$. T_k are linear projection functions that produce nine different attention weights for the sampling points per location, followed by a softmax function. The final $\text{head}_k^i \in \mathbb{R}^{d_v}$ in each head is the weighted average of the value vectors in $\mathbf{f}_i^v$ with attention weights. In this work, we employ $h = 8$ parallel attention heads and $d_v = d_f/h$.

Compared with deformable convolution that the sampling locations are learned without explicit supervision, the proposed box feature attention adopts fixed sampling points of the oriented boxes to obtain more precise features aligned with the objects. Different from AlignConv, we adopt the multihead attention method which generates attention weights for the sampling points in a pixelwise manner depending on the input query feature. In this way, we can obtain more accurate features for oriented objects by utilizing the spatial geometric information of oriented boxes.

C. Class-Aware Context Aggregation

In aerial scenarios, the foreground–background imbalance further increases the imbalance between the number of positive and negative samples, making it hard to learn a strong and unbiased classifier. To alleviate this problem, we design a lightweight CCA module to obtain more robust features for classifying the objects in oriented boxes into the right class. This module enhances the discrimination of intraclass features by suppressing a large amount of background and incorporates context information of foreground class. As shown in Fig. 4, our CCA module contains an auxiliary classification branch and a feature enhancement process. The auxiliary classification branch predicts a C-D vector for each location of the feature map that indicates the initial classification score. Based on this classification score, the network aggregates the class-aware context information to produce the enriched feature maps for final classification.

The details of CCA are shown in Fig. 7. Specifically, the input feature map $F \in \mathbb{R}^{H\times W\times 256}$ is first fed into the auxiliary classification branch to produce a coarse classification result.

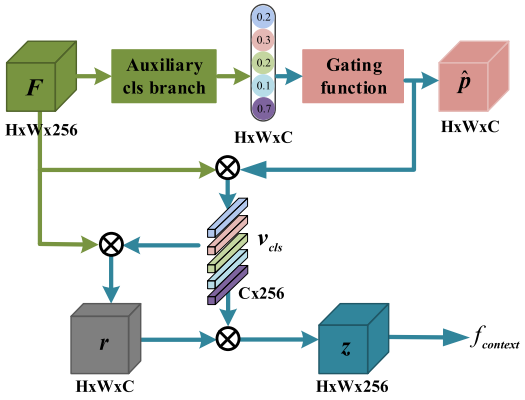


Fig. 7. Architecture details of class-aware context aggregation module. The input feature F is first fed into the auxiliary classification branch and then goes through a gating function to obtain the C-D category vector $\hat{p}$. Then, the class-specific feature representation v_{cls} is computed by aggregating all features on F weighted by the score map $\hat{p}$ of each class. After that, the soft relation vector r is computed by measuring the similarity between feature f and all the class representations v_{cls}. The class-aware context feature z is produced through aggregating all the class representations v_{cls} weighted by the relation vector r.

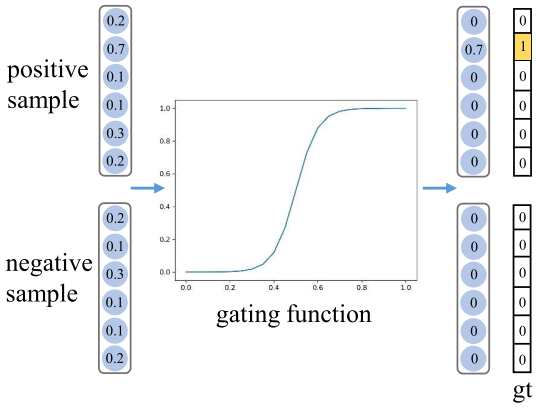


Fig. 8. Illustration of the gating function that suppresses the scores of negative sample and low confidence scores of positive sample.

$H \times W$ denotes the size of the feature map, and 256 is the number of feature channels. For each spatial location i over all the feature levels with features f_i, we obtain a C-D category vector $\hat{c}_i = \{p_{i,k}|k = 1, \ldots, C\}$. Since the coarse classification result is not as exact as the ground-truth class vector that may bring some noise, we feed it into a gating function to suppress the score of negative samples and the low confidence score of positive samples. The gating function is an approximate step function that is differentiable and can be described as

$$\begin{aligned}\hat{G}_{i,k} &= \frac{1}{1+e^{-A(p_{i,k}-t)}} \\ \hat{p}_{i,k} &= \hat{G}_{i,k}\cdot p_{i,k}\end{aligned} \tag{7}$$

where t is the predefined threshold and A indicates the amplifying factor. A is set as 20 empirically. Fig. 8 shows some examples and a plot of this gating function. With the newly produced classification score $\hat{p}_{i,k}$, we can get C score maps, including all the spatial locations across five feature levels, and each score map corresponds to a certain class. Then, a class-specific feature representation for each class is computed by

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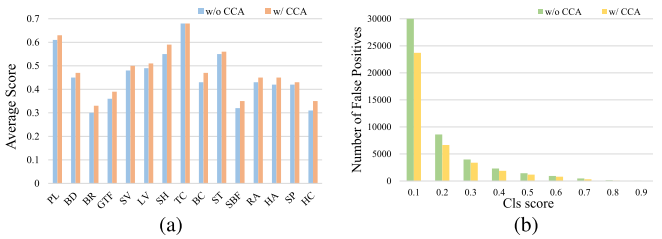


Fig. 9. (a) Improvement of the average score of each class on the DOTA1.0 validation set after applying the CCA module. (b) Histogram of the number of false positives at different classification score intervals on the DOTA1.0 validation set.

aggregating all features on a feature map weighted by the score map of this class, which is calculated as

$$v_k = \sum_i \hat{p}_{i,k} \cdot f_i \tag{8}$$

$$v_{\text{cls}} = \{v_k | k = 1, \ldots, C\}. \tag{9}$$

The confidence score $\hat{p}_{i,k}$ can reflect the estimated probability that a location i belongs to the kth class. By adaptively aggregating features that belong to the same class and discarding the background or other classes of irrelevant information, we expect the feature representation v_k to capture more intraclass context information. After that, to compute a context feature for each specific location, we mine the relationship between per location and each class. For the input feature f_i at location i with target class label c^*, a soft weight vector is computed via measuring the similarity between feature f_i and all the class representations v_{cls}, that is,

$$r_i = \varphi(f_i, v_{\text{cls}}) = f_i \odot v_{\text{cls}} = \{r_{i,k} | k = 1, \ldots, C\} \tag{10}$$

where φ represents the similarity estimation function and is implemented by the pointwise inner product. A softmax function is utilized to normalize the weights as

$$\hat{r}_{i,k} = \frac{e^{r_{i,k}}}{\sum_{k=1}^{C} e^{r_{i,k}}}. \tag{11}$$

We expect that f_i only captures the typical semantic feature of the target class. In other words, the similarity of f_i with other class representations should be discriminatively lower than with class c^*. After computing the similarity of features, the class-aware context feature z_i for location i is produced through aggregating all the class representations weighted by the relation vector, that is,

$$z_i = \psi\left(\sum_{k=1}^{C} \hat{r}_{i,k} \cdot \eta(v_k)\right) \tag{12}$$

where η is employed to project the feature f_i to an embedding space and ψ is a transformation function applied to increase feature nonlinearity since the weighted summation is a linear operation. These two functions are implemented by 1×1 convolution layer followed by batch normalization and ReLU. Our design fully utilizes relations between the features of each location and all the classes, reinforcing the representation z_i to exploit context information of target class.

As presented in Fig. 9, we visualize the average classification score of positive samples belonging to each class on the DOTA1.0 validation set. Fig. 2 shows the number of false positives at different classification score intervals. We can see that, after adding the CCA module, the average classification score of each class is increased, and the number of false positives is reduced across different score intervals.

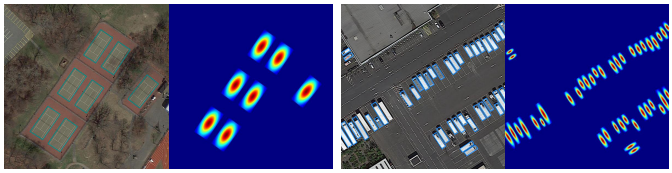


Fig. 10. Illustration of the 2-D Gaussian distribution for oriented boxes.

D. Soft Label Assignment

As mentioned above, the receptive fields of associated features at each location are not well aligned with the oriented boxes of diverse directions. Yang *et al.* [4] point out that the IoU between oriented boxes is sensitive to the angle changes, especially for objects with large aspect ratios. These factors increase the difficulty of oriented object detection, which makes that the assigned positive samples at different locations exhibit inconsistent localization ability to regress the oriented boxes. Therefore, it is more appropriate to introduce the dynamic label assignment strategy into oriented object detection instead of fixed label assignment. Inspired by the weighting mechanism in AutoAssign [31] for general object detection in natural images, we propose an SLA strategy that is more suitable for oriented object detection in aerial images.

In practice, we assign a soft weight to each sample's loss in order to consider the samples at different locations distinctly. To model this component, we first introduce a quality term that the high-quality samples refer to the samples that can produce more accurate predictions while low-quality samples and vice versa. The soft weights should allow to put the focus more on high-quality samples and reduce the focus on lower quality samples. Through this process, we can assign more appropriate labels to the samples and pay more attention to high-quality positive ones during training. More specifically, for an input image, suppose that there are N ground-truth boxes denoted as $\{B_n\}$, where $B_n = (c^{(n)}, x_{\text{ctr}}^{(n)}, y_{\text{ctr}}^{(n)}, w^{(n)}, h^{(n)}, \theta^{(n)})$ indicates the corresponding labels of the nth box, respectively. All the locations across five feature levels are represented as a set S. For each ground-truth box B_n, we assign those locations inside the box B_n across five levels as positive samples denoted as S_n. For each sample i, we can obtain a predicted box $\hat{B}_i$ and a classification output $\hat{c}_i = \{p_{i,k} | k = 1, \ldots, C\}$. $p_{i,k}$ represents the probability of belonging to class k and is normalized to $[0, 1]$ using a sigmoid function. We argue that the quality of a sample should consider both the prior spatial information and its ability to generate high-quality predictions. Moreover, since the object detection task needs to regress the precise box and classify it into the right class, the quality of predictions is related to both classification and localization accuracy [36]. Hence, we define the quality of a sample by taking into account prior spatial information, classification

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score, and localization precision simultaneously. The quality $Q_{i,n}$ of a sample i located at (x, y) in terms of predicting instance n is formulated as

$$Q_{i,n} = p^{\alpha}_{i,c^{(n)}} \cdot (\mathrm{IoU}(\hat{B}_i, B_n))^{(1-\alpha)} \cdot G_n(\mathbf{x}). \tag{13}$$

$p_{i,c^{(n)}}$ denotes the probability of sample i belonging to the ground-truth class of the nth instance box. $\mathrm{IoU}(\hat{B}_i, B_n)$ represents the IoU between the predicted OBB $\hat{B}_i$ and ground-truth OBB B_n, which indicates the localization accuracy of predictions. The hyperparameter α ranging in [0, 1] is used to control the weights of classification and regression. $\mathbf{x}$ denotes the coordinate (x, y), and $G_n(\mathbf{x})$ denotes a Gaussian distribution that reflects the prior spatial location of sample i for the oriented instance n. Specifically, given the OBB B_n denoted as $(c^{(n)}, x^{(n)}_{\mathrm{ctr}}, y^{(n)}_{\mathrm{ctr}}, w^{(n)}, h^{(n)}, \theta^{(n)})$, we define a 2-D Gaussian distribution. First, the probability density function of a general 2-D Gaussian distribution is given by

$$G_n(\mathbf{x}) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)}{2\pi\,|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \tag{14}$$

where μ and Σ denote the mean vector and the covariance matrix of the Gaussian distribution, respectively. Then, the OBB can be modeled into a 2-D Gaussian distribution $\mathcal{N}(\mu, \Sigma)$, where the density contour is the inscribed ellipse of the OBB. Here,

$$\begin{aligned} \mu &= \left(x^{(n)}_{ctr}, y^{(n)}_{\mathrm{ctr}}\right), \quad \Sigma = \mathbf{R}\Lambda\mathbf{R}^{\top} \\ \mathbf{R} &= \begin{bmatrix} \cos\theta^{(n)} & -\sin\theta^{(n)} \\ \sin\theta^{(n)} & \cos\theta^{(n)} \end{bmatrix} \\ \Lambda &= \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} = \begin{bmatrix} \left(\frac{w^{(n)}}{2}\right)^2 & 0 \\ 0 & \left(\frac{h^{(n)}}{2}\right)^2 \end{bmatrix} \end{aligned} \tag{15}$$

where $\mathbf{R}$ represents the rotation matrix and Λ represents the diagonal matrix of eigenvalues. The eigenvalues λ_1 and λ_2 are square of the semiaxes of the ellipse, respectively. We set the values of $G_n(\mathbf{x})$ for locations outside the inscribed ellipse of OBB as 0. The Gaussian distribution assigns a high value to locations close to the object's center while a low value to those away from the center. Fig. 10 shows a few examples of the Gaussian distribution.

Based on the quality $Q_{i,n}$ of samples, we can further derive the positive weight of sample i to instance n denoted as $w^{+}_{i,n}$. For samples in S^{+}_{n}, $w^{+}_{i,n}$ is formulated as

$$w^{+}_{i,n} = \frac{Q_{i,n} - \min_{i\in S^{+}_n}(Q_{i,n})}{\max_{i\in S^{+}_n}(Q_{i,n}) - \min_{i\in S^{+}_n}(Q_{i,n})} \tag{16}$$

$w^{+}_{i,n}$ is the normalized quality of samples by the peak value of samples in S^{+}_{n} to ensure at least one positive sample for object n. For samples not in S^{+}_{n}, that is, outside the bounding box B_n, $w^{+}_{i,n}$ is set to 0. Similarly, we need to assign a negative weight to each sample indicating the possibility of being set as a negative sample. Due to the correlation between positive and negative samples, i.e., the closer to positive, the farther to negative, we compute the negative weight for sample i as

$$w^{-}_{i} = 1 - \max_{n}\left(w^{+}_{i,n}\right). \tag{17}$$

$\max_n(w^{+}_{i,n})$ denotes the max positive weight of sample i for all ground-truth boxes. Therefore, the negative weights for locations outside all the ground-truth boxes are equal to 1, and the positive weights are equal to 0. They only participate in classification.

E. Detection Model

Adding the above three components to the baseline network, we obtain the framework of FSDet, as illustrated in Fig. 4. In the classification subnet, the class-aware context feature from the CCA module is concatenated with the feature map. Then, we employ a 1×1 convolutional layer to make the channel dimension consistent with the original feature.

During the training phase, the whole network is supervised by a multitask loss function, including the loss of the baseline network, as well as the auxiliary classification and regression branch. We adopt the outputs from the baseline network to compute the weights in SLA. Then, the loss function for the baseline network is formulated as

$$\begin{aligned} L_{\mathrm{base}} = &\frac{1}{N_{\mathrm{pos}}}\sum_i\left(\sum_n w^{+}_{i,n}L^{+}_{\mathrm{cls}}\left(\hat{c}_i, c^{(n)*}_i\right) + w^{-}_i L^{-}_{\mathrm{cls}}(\hat{c}_i, 0)\right) \\ &+\frac{\lambda_1}{N_{\mathrm{pos}}}\sum_i\sum_n w^{+}_{i,n}\left(L_{\mathrm{reg}}\left(\hat{d}_i, d^{(n)*}_i\right) + L_{\mathrm{IoU}}\left(\hat{q}_i, q^{(n)*}_i\right)\right) \end{aligned} \tag{18}$$

where $c^{(n)*}_i$, $d^{(n)*}_i$, and $q^{(n)*}_i$ are the targets of location i for ground-truth instance n. The loss function for auxiliary branches is given by

$$\begin{aligned} L_{\mathrm{aux}} = &\frac{1}{N_{\mathrm{pos}}}\sum_i\left(\sum_n w^{+}_{i,n}L^{+}_{\mathrm{cls}}\left(c'_i, c^{(n)*}_i\right) + w^{-}_i L^{-}_{\mathrm{cls}}\left(c'_i, 0\right)\right) \\ &+\frac{\lambda_1}{N_{\mathrm{pos}}}\sum_i\sum_n w^{+}_{i,n}L_{\mathrm{reg}}\left(d'_i, d^{(n)*}_i\right) \end{aligned} \tag{19}$$

where c'_i and d'_i denote the predictions of auxiliary classification and regression branch, respectively. The total training loss is calculated as

$$L_{\mathrm{total}} = L_{\mathrm{base}} + L_{\mathrm{aux}}. \tag{20}$$

We summarize the entire training process in Algorithm 1.

During the inference phase, the input image is passed through the whole network, and the outputs from auxiliary branches are only used to refine the corresponding feature maps. We multiply the classification score by the predicted IoU and adopt a postprocessing NMS to produce the final detections.

IV. Experiments

A. Experimental Setting

1) Datasets: We conduct experiments on several datasets for object detection in aerial images, including DOTA1.0 [37], DOTA1.5, HRSC2016 [38], and UCAS-AOD [39]. Objects in all these three datasets are annotated with OBBs. DOTA1.0 is a large-scale publicly available dataset for aerial object detection, which contains 2806 images in total, with 15 different categories and 188 282 annotated instances. DOTA1.5 contains

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Algorithm 1 Detection Model

Input: The classification features $\mathscr{F}^{cls}$ and regression features $\mathscr{F}^{reg}$ accross L pyramid levels of an input image, the ground-truth oriented boxes
Output: Computed loss for network training

for *each* $F^{cls} \in \mathscr{F}^{cls}$, $F^{reg} \in \mathscr{F}^{reg}$ **do**
  Generate a set of oriented boxes $B = \{B_i\}_{i=1}^{N_l}$ and class logits $\hat{C} = \{\hat{c}_i\}_{i=1}^{N_l}$;
  `// Oriented Feature Refinement`
  for $i = 1$ *to* N_l **do**
    Calculate the enhanced feature $\hat{F}^{cls}$ and $\hat{F}^{reg}$ at each location,
    $\hat{F}_i^{cls} \leftarrow OFR(B_i, F^{cls})$;
    $\hat{F}_i^{reg} \leftarrow OFR(B_i, F^{reg})$;
  end
  $\hat{F}_f^{cls} \leftarrow F^{cls}.reshape(N_l, D)$;
end
$F_{flatten}^{cls} \leftarrow Concat(\hat{F}_f^{cls})$;
`// Class-aware Context Aggregation`
for $i = 1$ *to* $\sum_{l=1}^{L} N_l$ **do**
  Calculate the context representation according to Eq. 12;
end
Split the context representation to L pyramid levels, get $F_{context}^{cls}$;
for *each* $\hat{F}^{cls} \in \hat{\mathscr{F}}^{cls}$, $\hat{F}^{reg} \in \hat{\mathscr{F}}^{reg}$, $F_{context}^{cls} \in \mathscr{F}_{context}^{cls}$ **do**
  $F_{final}^{cls} \leftarrow conv(concat(\hat{F}^{cls}, F_{context}^{cls})$;
  get final class logits using F_{final}^{cls} and $\hat{F}^{reg}$;
end
`// Soft label Assignment`
for $n = 1; n < N_g; n++$ **do**
  for $i = 1$ *to* $\sum_{l=1}^{L} N_l$ **do**
    Calculate the positive weight according to Eq. 16;
  end
  Calculate the negative weight according to Eq. 17;
end
Compute the total loss according to Eq. 20;
return loss

402 089 instances, which adds more small instances less than ten pixels and a new category Container Crane based on DOTA1.0. The images come from multiple sensors and platforms covering a variety of scenes with sizes ranging from 800 × 800 to 4000 × 4000. We utilize the standard training, validation, and testing splits with 1411, 458, and 937 images, respectively. Following the setting in [40], the images are divided into a fixed size of 800 × 800 with a stride of 200 pixels. HRSC2016 is a high-resolution ship detection dataset that includes 436 images for training, 181 images for validation, and 444 images for testing. The size of the images ranges from 300 × 300 to 1500 × 900. We use the training and validation sets to train our model and evaluate the model on the testing set. UCAS-AOD is commonly used for aerial car and aircraft detection, which includes 1510 images. For a fair comparison, we follow the same dataset splits as in [40], and the ratio between the training set, the validation set, and the testing set is 5:2:3.

TABLE I
ABLATION EXPERIMENTS WITH DIFFERENT COMPONENTS ON THE DOTA1.0 DATASET

Rot	IoU perdiction	SLA	OFR	CCA	mAP(%)
					73.21
	√				73.54 (+0.45%)
√	√				74.69 (+2.02%)
√	√	√			76.08 (+3.92%)
√	√	√	√		77.46 (+5.80%)
√	√	√		√	76.93 (+5.08%)
√	√	√	√	√	**77.85 (+6.33%)**

2) Implementation Details: We adopt ResNet50 [41] pre-trained on ImageNet [42] as the backbone network for all the experiments unless otherwise specified. In all experiments, we employ the stochastic gradient descent (SGD) optimizer over four GPUs with a total of eight images per minibatch. The weight decay and momentum are set to 0.0001 and 0.9, respectively. The initial learning rate is set to 0.01 and divided by 10 at every learning rate decay. All the images are resized to 800 × 800 in both training and inference. For data augmentation, random flip and random rotation are adopted during training. For experiments on DOTA1.0 and DOTA1.5 datasets, models are trained for 36 epochs, and the learning rate decays at {24, 33} epochs. We train the models on the HRSC2016 dataset for 5.5k steps, and the learning rate decays at {3.7k, 5k} steps. As for the UCAS-AOD dataset, models are trained for 6.8k steps, and the learning rate decays at {4.5k, 6.2k}. The running time is measured on a single 2080ti GPU with 11-GB memory.

B. Ablation Study

In this section, we conduct a set of ablation experiments on the DOTA1.0 dataset to analyze each proposed module and some important hyperparameters in detail. For a fair comparison, we use the same parameter settings for all the experiments except for specified changes to the components. The result of ablation experiments is shown in Table I, and the first row represents the result of the original FCOS with a rotational head for detecting arbitrary-oriented objects. Then, we replace the centerness branch with the IoU prediction branch (i.e., FCOS-R) to predict the IoU between the regressed box and ground-truth box. Since IoU is more consistent with the evaluation metric, we obtain a slight improvement of 0.33%. After employing random rotation for data augmentation, the performance of the baseline FCOS-R network reaches 74.69% mAP. Based on this network, we add each module progressively to validate their effectiveness.

1) Effectiveness of Soft Label Assignment: First, we conduct experiments to validate the effectiveness of the SLA strategy. Specifically, we add the SLA to the baseline network FCOS-R. As shown in rows 3 and 4 of Table I, the network trained with SLA strategy achieves an improvement of 1.39% mAP, which demonstrates the superiority of SLA. As stated in the introduction, samples at different locations exhibit inconsistent ability

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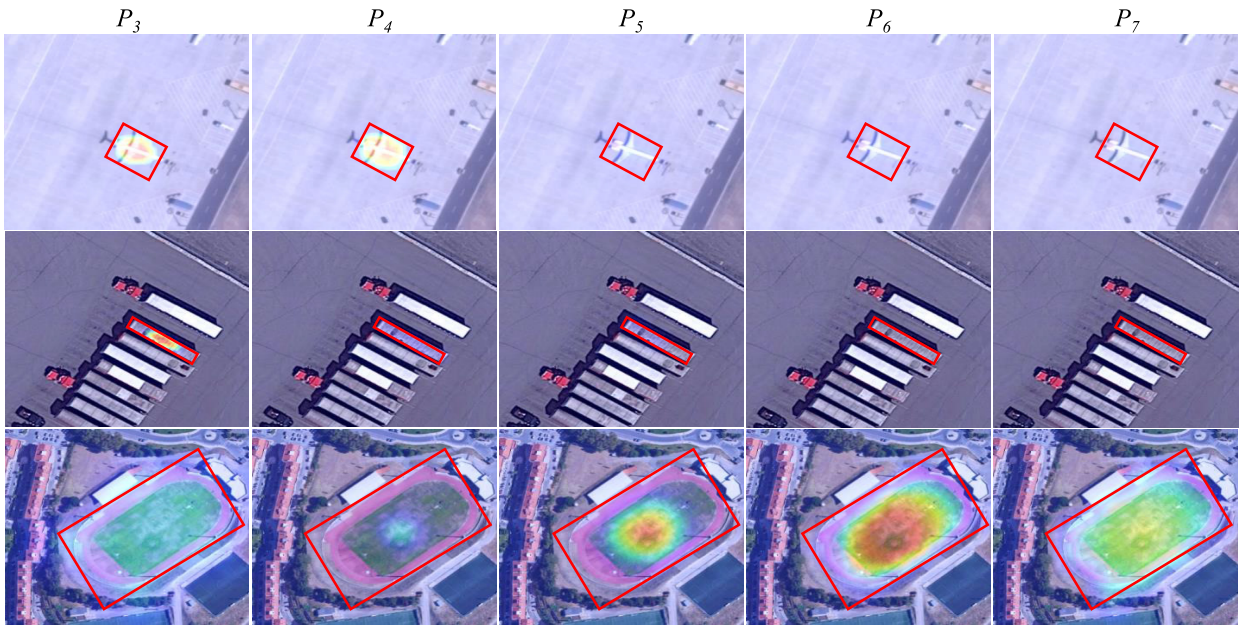


Fig. 11. Visualizations of positive weight maps at different feature levels.

to regress the oriented boxes, and the results indicate that it is beneficial to assign soft weights to these samples adaptively according to their quality during the training procedure.

To further comprehend the effect of SLA, we visualize the positive weights of a few sample images at different feature levels in Fig. 11. The weight map is computed for the specific object surrounded by the oriented box in the image. We can observe that the objects can be assigned to multiple feature levels and spatial locations adaptively with soft weights. Meanwhile, the weight maps are also consistent with the concept of FPN that small objects get a stronger response in the weight map of a low feature level with more detailed information, while large objects get a stronger response at a high feature level with a larger receptive field. On the other hand, the weight map on the high-response feature level is well aligned with the object, i.e., the regions close to the object's center tend to acquire high attention, and those regions close to the boundary receive low attention. These indicate that the proposed SLA can focus on appropriate feature scales and more informative spatial locations while paying less attention to other irrelevant regions.

2) Effectiveness of Oriented Feature Refinement Module: Since the SLA is only used during training and has no additional cost for inference, we adopt this strategy for training the network in subsequent experiments by default. We explore the influence of the OFR module by adding the box feature attention mechanism to the classification and regression subnet of the baseline network, respectively. We also compare the proposed OFR with other feature refinement methods, such as deformable convolution and alignment convolution. The results are shown in Table II. It can be seen that AlignConv achieves better performance than DeformConv, and the proposed OFR outperforms AlignConv while introducing fewer parameters and computation. Different from DeformConv that learns the sampling locations with no explicit supervision, we employ the nine fixed sampling points of the oriented box. Compared with AlignConv that shares the same convolutional kernel across the entire feature map, our OFR can produce adaptive attention weights for the samples at different spatial locations. Moreover, we can see that adding the box feature attention mechanism to the classification and regression subnet separately brings a performance improvement; thus, we employ this in both two subnets. The results demonstrate the effectiveness of the proposed OFR for extracting features from oriented objects.

TABLE II
EVALUATION WITH DIFFERENT CONVOLUTIONS IN THE OFR MODULE

Feature Refinement	Cls	Reg	Backbone	Parameters	GFlops	mAP(%)
			ResNet50	31.92M	206.86	76.08
Deformable Convolution	√		ResNet50	31.98M	208.24	76.32
		√	ResNet50	31.98M	208.24	76.69
	√	√	ResNet50	32.05M	209.61	76.81
Align Convolution	√		ResNet50	32.58M	221.38	76.61
		√	ResNet50	32.58M	221.38	76.93
	√	√	ResNet50	32.65M	222.75	77.24
OFR	√		ResNet50	32.52M	220.19	76.82
		√	ResNet50	32.52M	220.19	77.13
	√	√	ResNet50	32.53M	220.38	**77.46**

Furthermore, we visualize the classification and regression feature map of a few samples in Fig. 12. We can see that, by introducing the OFR module, the high-response area of classification is more likely to focus on the objects, and the attention of regression is more distributed on the edge of the objects.

3) Effectiveness of Class-Aware Context Aggregation Module: We explore the influence of the CCA module by adding it to the baseline network. We find that FSDet equipped with CCA achieves a 0.85% mAP improvement compared with the baseline network. Besides, we count the number

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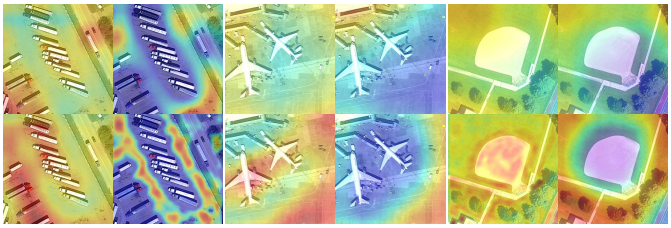


Fig. 12. Visualization results of the feature maps for classification and regression with the OFR module. (Top row) Results of the baseline network. (Bottom row) Results after adding the OFR module. The left of each pair of images denotes the features for classification, while the right is the features for regression.

TABLE III

EVALUATION WITH THE CCA MODULE IN TERMS OF FALSE POSITIVES

Category	CCA	Recall(%)									
		10	20	30	40	50	60	70	80	90	95
BD		0	1	3	5	8	13	28	76	670	670
	√	**0**	**1**	**3**	**3**	**4**	**7**	**11**	**25**	**229**	**229**
GTF		2	4	10	20	26	38	66	220	995	995
	√	**0**	**1**	**5**	**7**	**12**	**17**	**27**	**68**	**409**	**409**
BC		0	0	1	2	10	32	171	478	478	478
	√	**0**	**0**	**0**	**2**	**6**	**15**	**27**	**410**	**410**	**410**
HC		1	2	3	5	19	42	203	572	572	572
	√	**0**	**0**	**2**	**3**	**11**	**36**	**147**	**434**	**434**	**434**

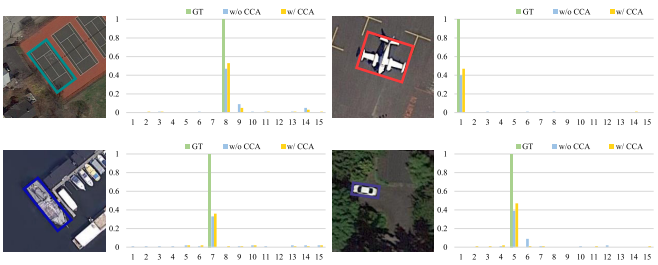


Fig. 13. Effect of the CCA module on the classification score of samples.

of false positives under different recall rates of four categories with significant improvement in Table III. As can be observed, our CCA effectively reduces the false positives across different recall rates, demonstrating the effectiveness of the CCA module. Fig. 13 shows the classification scores of some examples. We can see that, after adding the CCA module, the score for the corresponding class label increases, and others are suppressed. We show the qualitative comparison of detection results on a few images in Fig. 14. It indicates that the model trained with the CCA module can reduce some false positive detections and classify the objects in boxes into the right classes. Moreover, we visualize the learned feature representations of each class in Fig. 15. After applying the CCA module, the features belonging to the same class are more compact, and the features of different classes are more separated. It suggests that the proposed CCA is conducive to exploiting the useful intraclass context information and reducing the noise of other regions, which is beneficial to obtain more accurate classification and reduce the false positives.

Finally, by including all the proposed OFR, CCA, and SLA, our FSDet achieves an mAP of 77.85%, increasing the mAP by 3.16% compared with the baseline FCOS-R, indicating that these three components are compatible with each other well and can effectively improve the detection performance.

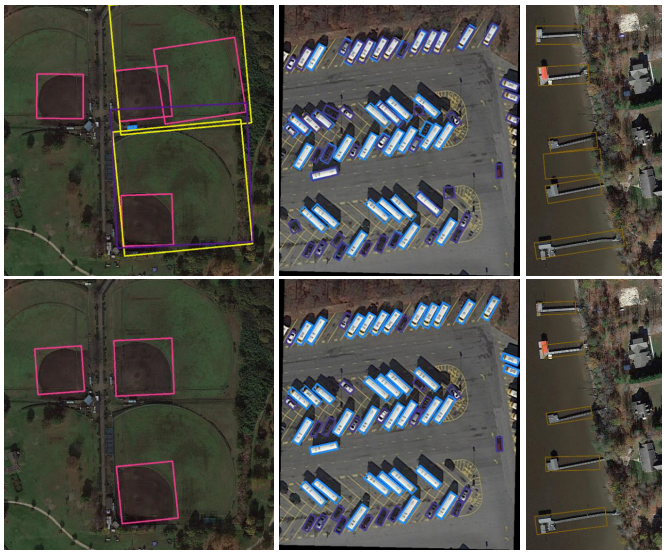


Fig. 14. Effect of the CCA module on detection results. (Top row) Results of the baseline network. (Bottom row) Results after adding the CCA module.

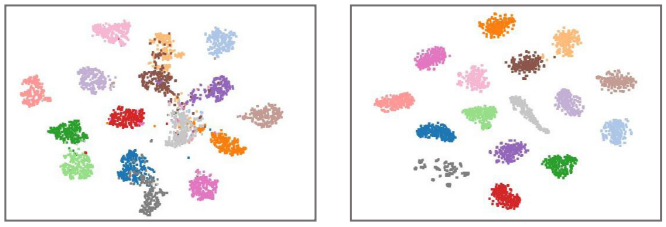


Fig. 15. t-SNE visualization of feature embeddings of each class learned with and without the CCA module.

TABLE IV

COMPLEXITY ANALYSIS

Methods	Backbone	Parameters	GFlops	FPS	mAP(%)
FCOS-R (w SLA)	ResNet50	31.92M	206.86	15.7	76.08
+OFR	ResNet50	32.53M	220.38	13.2	77.46
+CCA	ResNet50	33.66M	227.09	12.6	76.93
FSDet	ResNet50	34.26M	240.61	11.3	77.85

TABLE V

EVALUATION OF USING VARYING t IN THE GATING FUNCTION

t	0.2	0.3	0.5	0.7	0.9
mAP(%)	76.41	76.72	**76.93**	76.64	76.23

TABLE VI

EVALUATION OF USING VARYING α IN SLA

α	0.0	0.2	0.4	0.6	0.7	0.8	0.9	1.0
mAP(%)	75.23	75.46	75.68	75.82	75.97	**76.08**	75.62	75.12

4) Complexity Analysis: We analyze the complexity and detection performance of the proposed method in Table IV, including the model parameters, flops, speed, and mAP. The

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TABLE VII
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA1.0 DATASET

Methods	Backbone	AF	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP(%)
Two-stage Methods																		
FR-O [37]	ResNet101		79.09	69.12	17.17	63.49	34.20	37.16	36.20	89.19	69.60	58.96	49.40	52.52	46.69	44.80	46.30	52.93
R-DFPN [43]	ResNet101		80.92	65.82	33.77	58.94	55.77	50.94	54.78	90.33	66.34	68.66	48.73	51.76	55.10	51.32	35.88	57.94
R^2CNN [44]	ResNet101		80.94	65.67	35.34	67.44	59.92	50.91	55.81	90.67	66.92	72.39	55.06	52.23	55.14	53.35	48.22	60.67
RRPN [45]	ResNet101		88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58	61.01
ICN [46]	ResNet101		81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
CSL [47]	ResNet50		89.09	79.22	45.78	67.94	66.58	56.29	71.55	90.80	80.50	76.61	58.53	61.01	59.92	69.51	57.59	68.73
RoI-Transformer [2]	ResNet101		88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
CAD-Net [3]	ResNet101		87.80	82.40	49.40	73.50	71.10	63.50	76.70	**90.90**	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
SCRDet [22]	ResNet50		89.48	79.83	50.43	62.51	76.72	74.31	85.42	90.86	84.19	82.51	42.84	60.15	74.18	70.62	51.81	71.72
Gliding vertex [5]	ResNet101		89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
ReDet [48]	ReResNet50		88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
Oriented R-CNN [49]	ResNet101		88.86	83.48	55.27	76.92	74.27	82.10	87.52	**90.90**	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
One-stage Methods																		
PIoU [50]	ResNet101		80.90	69.70	24.10	60.20	38.30	64.40	64.80	**90.90**	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
CFC-Net [40]	ResNet101		89.08	80.41	52.41	70.02	70.02	78.11	87.21	90.89	84.47	85.64	60.51	61.52	67.82	68.02	50.09	73.50
R^3Det [4]	ResNet101		88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
S^2A-Net [20]	ResNet50		89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
R^3Det-GWD [51]	ResNet50		88.82	82.94	55.63	72.75	78.52	83.10	87.46	90.21	86.36	85.44	64.70	61.41	73.46	76.94	57.38	76.34
R^3Det-KLD [52]	ResNet50		88.90	84.17	**55.80**	69.35	78.72	**84.08**	87.00	89.75	84.23	85.73	64.74	61.80	**76.72**	78.49	**70.89**	77.36
IENet [12]	ResNet101	√	80.20	64.54	39.82	32.07	49.71	65.01	52.58	81.45	44.66	78.51	46.54	56.73	64.40	64.24	36.75	57.14
Axis Learning [13]	ResNet101	√	79.53	77.15	38.59	61.15	67.53	70.49	76.30	89.66	79.07	83.53	47.27	61.01	56.28	66.06	36.05	65.98
DRN [53]	ResNet101	√	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
O^2-DNet [54]	ResNet101	√	89.31	82.14	47.33	61.21	71.32	74.03	78.62	90.76	82.23	81.36	60.93	60.17	58.21	66.98	61.03	71.04
BBAVectors [14]	ResNet101	√	88.35	79.96	50.69	62.18	78.43	78.98	87.94	90.85	83.58	84.35	54.13	60.24	65.22	64.28	55.70	72.32
GGHL [32]	DarkNet53	√	**89.74**	**85.63**	44.50	**77.48**	76.72	80.45	86.16	90.83	**88.18**	**86.25**	67.07	69.40	73.38	68.45	70.14	76.95
FCOSR [33]	ResNeXt50-32×4	√	88.88	82.68	50.10	71.34	**81.09**	77.40	**88.32**	90.80	86.03	85.23	61.32	68.07	75.19	**80.37**	70.48	77.15
FSDet(ours)	ResNet50	√	89.20	84.33	55.38	74.81	80.85	78.79	88.24	90.85	86.85	85.85	**67.14**	64.79	76.49	75.11	69.02	**77.85**

SLA strategy is only adopted for calculating the weights of samples and used in loss function during training; hence, it brings no extra cost for inference. We can see that only adding the SLA to the baseline network achieves a fast speed and high detection accuracy. After applying two feature enhancement modules, the detection accuracy obtains an improvement than the baseline network while introducing a few more parameters and computations that mainly come from the auxiliary classification and regression branch.

5) Evaluation of Hyperparameters:

a) Choice of the threshold t: The hyperparameter t is introduced in the gating function to adjust the threshold that differentiates regions of a certain class from the background and other classes. We conduct experiments to find the choice of t. The result is presented in Table V. As t increases, the ability to suppress noisy areas is stronger. With $t = 0.5$, the proposed method achieves the best result of 76.93% in terms of mAP. However, when t is too large, the performance drops. We conjecture that the possible reason is that the coarse classification score is not accurate enough, and the increase in t may also suppress the information of potential intraclass regions, resulting in performance degradation. Thus, we choose $t = 0.5$ in our experiments.

b) Choice of the factor α: When determining the quality of samples, the factor α is used to adjust the ratio of classification confidence and localization accuracy. We conduct experiments to find the choice of α, and the result is shown in Table VI. As α increases, the quality of a sample is more affected by IoU between the predicted box and ground truth, i.e., the localization accuracy, and the performance obtains continually improvement. However, when $\alpha = 1.0$ where we only maintain the contribution of localization performance and ignore classification accuracy, the performance drops a lot. This indicates that the overall quality of samples is related to both classification and localization, and thus, they should be considered simultaneously. Finally, we choose $\alpha = 0.8$ in this article.

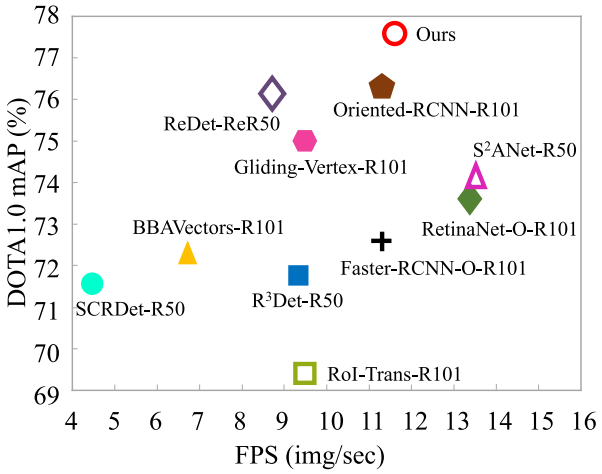


Fig. 16. Speed versus accuracy on the DOTA1.0 dataset.

C. Comparison With the State of the Art

To evaluate the effectiveness of the proposed FSDet, we conduct extensive experiments compared with the state-of-the-art methods on three public aerial object detection datasets, including DOTA1.0, DOTA1.5, HRSC2016, and UCAS-AOD.

1) DOTA1.0 Dataset: DOTA is a more challenging dataset that contains lots of arbitrary-oriented objects of different categories in various scenarios. The comparison results are depicted in Table VII. Moreover, we test the speed and accuracy of different methods under the same setting, as illustrated in Fig. 16. As can be seen that our method obtains the best performance of 77.85%, which outperforms all compared methods, the results show that anchor-free detectors have great potential to achieve state-of-the-art detection accuracy

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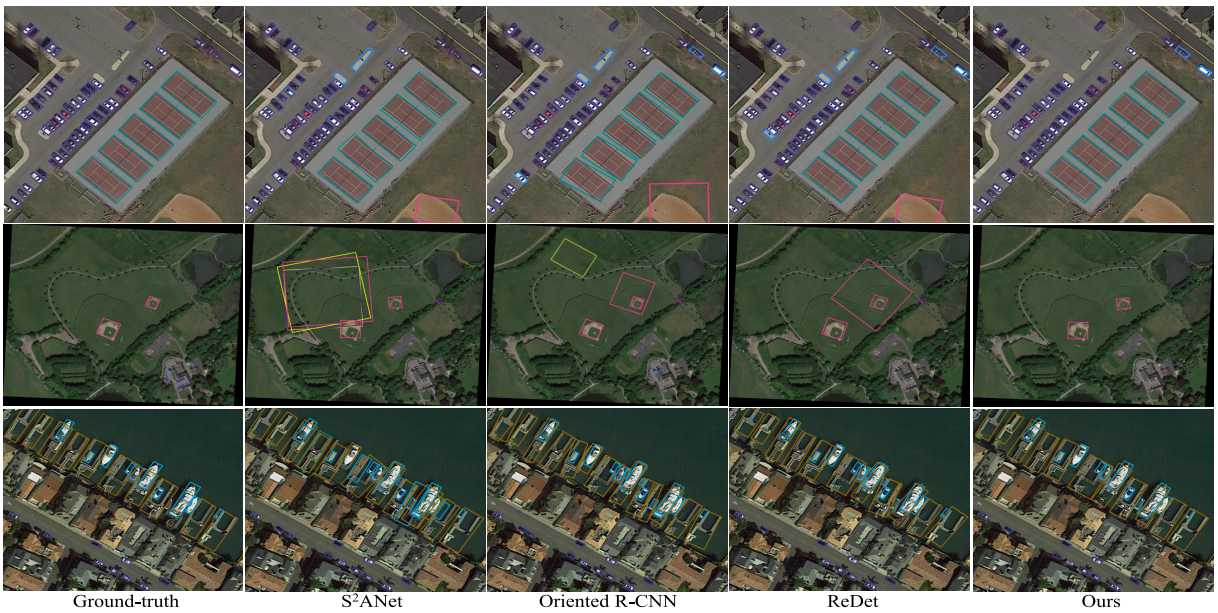


Fig. 17. Visualization results of different methods on the DOTA1.0 dataset.

TABLE VIII
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA1.5 DATASET

Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP(%)
RetinaNet-O [55]	ResNet50	71.43	77.64	42.12	64.65	44.53	56.79	73.31	**90.84**	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
FR-O [37]	ResNet50	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
Mask R-CNN [56]	ResNet50	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67
HTC [57]	ResNet50	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
ReDet [48]	ReResNet50	79.20	**82.81**	**51.92**	71.41	52.38	**75.73**	80.92	90.83	75.81	68.64	49.29	**72.03**	73.36	70.55	63.33	11.53	66.86
FSDet(ours)	ResNet50	**80.81**	81.22	51.35	**72.89**	**54.53**	74.12	**81.07**	90.81	**78.40**	**76.85**	**52.81**	70.42	**73.73**	**71.89**	**68.45**	**14.67**	**68.37**

as anchor-based detectors in aerial images. In this work, only single-scale images are employed for training and testing. We do not use the tricks such as multiscale training and testing, as well as a larger detection backbone. Fig. 17 shows a qualitative comparison of detection results of different methods. We find that our FSDet can reduce some false detections and produce more accurate results. Fig. 18 shows some detection results of our FSDet on DOTA. We observe that our FSDet performs well in various scenarios in terms of objects with different scales and appearance changes. For the first row where the vehicles in images are densely arranged, FSDet can locate these closely jointed instances and separate them accurately. Moreover, for objects with large aspect ratios, such as harbors and bridges, our method can also detect them precisely. It suggests that FSDet can deal with all these challenging cases and show promising results, demonstrating the robustness of the proposed method.

2) DOTA1.5 Dataset: For experiments on the DOTA1.5 dataset, we show the results of a few comparison methods that have evaluated on this dataset, such as RetinaNet-O [55], FR-O [37], Mask R-CNN [56], HTC [57], ReDet [48], and our FSDet in Table VIII. We can see that the previous ReDet achieves the leading performance of 66.86% among the comparison methods, and our FSDet obtains an improvement of 1.51% and achieves the best performance of 68.37%. This demonstrates the effectiveness of our FSDet for detecting the objects in aerial images, which benefits from the complementary effect of the proposed components.

TABLE IX
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET

Methods	Backbone	NA	mAP(%)	Runtime(s)
Anchor-based Methods				
R^2CNN [44]	ResNet101	21	73.07	0.5
RC1 & RC2 [11]	VGG16	-	75.7	>1
RRPN [45]	ResNet101	54	79.08	0.28
YOLOv3-R [58]	Darknet53	9	79.53	**0.03**
R^2PN [59]	VGG16	24	79.60	-
RetinaNet-H [4]	ResNet101	9	82.89	0.07
RRD [16]	VGG16	13	84.30	slow
RoI-Transformer [2]	ResNet101	5	86.20	0.11
Gliding Vertex [5]	ResNet101	5	88.20	-
RetinaNet-R [4]	ResNet101	121	89.20	0.10
R^3Det [4]	ResNet101	21	89.30	0.08
CFC-Net [40]	ResNet101	1	89.50	-
S^2A-Net [20]	ResNet101	1	90.17	-
ReDet [48]	ReResNet50	15	**90.46**	-
Anchor-free Methods				
IENet [12]	ResNet101	-	75.01	0.06
Axis Learning [13]	ResNet101	-	78.15	0.07
CenterNet-R [17]	ResNet101	-	83.72	0.12
BBAVectors [14]	ResNet101	-	88.60	0.09
Ours	ResNet101	-	90.38	0.09

3) HRSC2016 Dataset: We compare FSDet with two kinds of methods, namely, anchor-based methods and anchor-free methods on the HRSC2016 dataset. The detection results are shown in Table IX. We can observe that our FSDet

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Fig. 18. Visualization results of the proposed FSDet on the DOTA1.0 dataset.

TABLE X
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE UCAS-AOD DATASET

Methods	car	airplane	mAP(%)
YOLOv3-R [58]	74.63	89.52	82.08
RetinaNet-R [55]	84.64	90.51	87.57
FR-O [37]	86.87	89.86	88.36
RoI-Transformer [2]	88.02	90.02	89.02
CFC-Net [40]	89.29	88.69	89.49
DAL [30]	89.25	90.49	89.87
Ours	**89.94**	**90.68**	**90.31**



Fig. 19. Visualization results of the proposed FSDet on the HRSC2016 dataset.

achieves a competitive performance of 90.38% compared with other methods. Specifically, among anchor-based methods, ReDet shows a performance of 90.46% and outperforms the other methods. Among anchor-free methods,



Fig. 20. Visualization results of the proposed FSDet on the UCAS-AOD dataset.

BBAVectors [14] obtains the best result of 88.60%. The proposed FSDet achieves an improvement of 1.60% over BBAVectors. Although our FSDet is an anchor-free method without the assistance of presetting anchors, it can still achieve competitive performance with anchor-based detectors. Fig. 19 shows some detection results of the proposed method on HRSC2016. It can be seen that our FSDet can accurately detect arbitrary-oriented ships. Also, it exhibits outstanding performance when detecting ships with large aspect ratios and different sizes. These results indicate that FSDet is robust to arbitrary orientations, varying scales, and large aspect ratios that are prevalent for detecting objects in aerial scenarios.

4) UCAS-AOD Dataset: The experimental results on UCAS-AOD are shown in Table X. Again, we find that FSDet obtains the best performance of 90.31% compared with other

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methods. It indicates that the proposed method, including the specially designed SLA strategy and two feature enhancement modules, can assign more appropriate attention to samples at different locations and learn more robust features for oriented objects with rich context information, which are crucial for object detection in aerial images. Some detection results on UCAS-AOD are shown in Fig. 20.

V. Conclusion

In this article, we propose a new precise and efficient anchor-free network for arbitrary-oriented object detection in aerial images. The OFR module is designed to extract more effective features for oriented objects by utilizing the geometric information of oriented boxes. The CCA module is proposed to fully exploit the intraclass context information while suppressing the large amounts of noisy background and other irrelevant classes. Moreover, considering the shortcoming of inaccurate label assignment in current anchor-free detectors, we develop an SLA strategy to differentiate the samples with inconsistent localization quality, which is quite effective to promote detection performance. We conduct extensive experiments on three public datasets, including DOTA1.0, DOTA1.5, HRSC2016, and UCAS-AOD, and the results show that the proposed FSDet achieves state-of-the-art performance.

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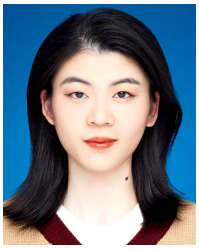
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